Little Tiger Invitational - Repaired practice edition

Division B | Thermodynamics | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

24. (1 points) Authored replacement for source item 24: For a closed system performing only pressure-volume boundary work, which process has zero boundary work because its volume is constant?

Answer: _____
_____
_____

30. (1 points) Authored replacement for source item 30: State the Kelvin-Planck form of the second law in your own words, without attributing it to the wrong scientist.

Answer: _____
_____
_____

32. (3 points) Authored replacement for source item 32: Calculate the energy required to heat 16.50 kg of silver from 20 C to its melting point and melt it completely. Added assumption: melting point is 962 C. Use c=230 J/(kg K), latent heat of fusion 88,000 J/kg, and no heat loss.

Answer: _____
_____
_____

repair-case. (4 points) Authored written substitute for unavailable source material: Why check delta U from temperature too?

Unit: Internal energy and the first law.
Worked-case data: A system receives 500 J heat and does 200 J work. Find delta U and chemistry work.

Answer: _____
_____
_____